

BLOG WEBSITE SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Name

Global Blog Website

Version

1.0

Technology Stack

Backend

- Laravel 12
- PHP 8.3+

Database

- MySQL 8+

Frontend

- HTML5
- CSS3
- Bootstrap 5.3+
- JavaScript (ES6+)
- Blade Template Engine

Server

- Apache / Nginx
- Linux VPS Hosting

1. Project Overview

Objective

Develop a modern, fast, secure, SEO-friendly blog website where administrators can publish articles and visitors can read, search, share, and interact with blog content.

The platform must be scalable, mobile responsive, and production-ready.

2. User Roles

Guest Visitor

Can:

- View blog posts
 - Search articles
 - Browse categories
 - Browse tags
 - Share articles
 - Submit comments
-

Registered User

Can:

- Login
 - Manage profile
 - Save favorite articles
 - Submit comments
 - Receive notifications
-

Author

Can:

- Create blog posts
 - Edit own posts
 - Upload images
 - Manage drafts
-

Administrator

Can:

- Manage all content
- Manage users

- Manage categories
 - Manage tags
 - Moderate comments
 - Manage settings
 - View analytics
-

3. Frontend Features

Home Page

Sections:

Hero Section

Display:

- Featured articles
- Latest articles

Trending Articles

Display popular posts based on:

- Views
- Comments

Categories Section

Display:

- Category image
- Category name
- Post count

Newsletter Subscription

Collect subscriber emails.

Blog Listing Page

Features:

- Pagination
- Search

- Category Filter
 - Tag Filter
 - Author Filter
 - Sort By Date
-

Single Blog Page

Display:

- Title
 - Featured Image
 - Author
 - Publish Date
 - Reading Time
 - Content
 - Tags
 - Categories
 - Social Share Buttons
 - Related Posts
-

Author Profile Page

Display:

- Author photo
 - Bio
 - Social links
 - Published articles
-

Contact Page

Fields:

- Name
 - Email
 - Subject
 - Message
-

About Us Page

Dynamic content managed by admin.

4. Blog Management Module

Create Blog Post

Fields:

Field	Type
Title	Text
Slug	Auto Generated
Category	Dropdown
Tags	Multiple
Featured Image	Upload
Short Description	Text
Content	Rich Editor
Meta Title	Text
Meta Description	Text
Canonical URL	Text
Status	Draft/Published
Publish Date	Datetime

Rich Text Editor

Support:

- Headings
- Paragraphs
- Lists
- Tables
- Images
- Hyperlinks
- Code Blocks
- Quotes

Recommended:

- CKEditor 5

- TinyMCE
-

Draft System

Status:

- Draft
 - Published
 - Scheduled
-

5. Category Management

Fields:

- Name
- Slug
- Description
- Category Image
- SEO Metadata

Functions:

- Add
 - Edit
 - Delete
-

6. Tag Management

Fields:

- Name
- Slug

Functions:

- Add
 - Edit
 - Delete
-

7. Comment System

Features:

- Nested comments
- Reply comments
- Comment moderation
- Approve/Reject comments

Security:

- Google reCAPTCHA
 - Spam protection
-

8. Search System

Search Features

Search By:

- Title
- Content
- Category
- Tags
- Author

Implementation:

- MySQL Full Text Search

Future Upgrade:

- Elasticsearch
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9. Media Library

Supported Formats

Images:

- JPG
- PNG

- WebP
- SVG

Documents:

- PDF

Functions:

- Upload
 - Delete
 - Replace
 - Organize
-

10. User Management

User Fields

- Name
- Email
- Password
- Profile Image
- Bio
- Status

Roles:

- User
 - Author
 - Admin
-

11. Authentication Module

Features:

- Registration
- Login
- Logout
- Forgot Password
- Reset Password
- Email Verification

Laravel Features:

- Laravel Breeze
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12. SEO Requirements

Technical SEO

Automatic Generation:

- XML Sitemap
 - Robots.txt
 - RSS Feed
-

Metadata

Support:

- Meta Title
 - Meta Description
 - Keywords
 - Canonical URL
-

Social Sharing

Support:

- Open Graph
 - Twitter Card
-

Structured Data

Schema Types:

- BlogPosting
 - Article
 - BreadcrumbList
-

13. Performance Requirements

Target Performance

Homepage Load Time:

< 2 Seconds

Article Page:

< 2 Seconds

Optimization

- Image Compression
 - Lazy Loading
 - Browser Caching
 - Gzip Compression
 - Minified CSS
 - Minified JavaScript
-

14. Security Requirements

Application Security

Protection Against:

- SQL Injection
- XSS
- CSRF
- Brute Force Attacks

Laravel Security:

- CSRF Tokens
 - Eloquent ORM
 - Input Validation
-

Authentication Security

- Password Hashing
 - Rate Limiting
 - Session Timeout
-

15. Analytics

Dashboard Metrics:

- Total Posts
- Total Users
- Total Views
- Total Comments
- Popular Posts

Integration:

- Google Analytics 4
 - Google Search Console
-

16. Newsletter Module

Features:

- Email Subscription
- Subscriber Management
- Export Subscribers

Fields:

- Name
 - Email
-

17. Advertisement Management

Admin Can:

- Add Google AdSense Code
- Add Banner Ads
- Manage Ad Positions

Positions:

- Header
 - Sidebar
 - In Article
 - Footer
-

18. Admin Dashboard

Modules:

Dashboard

- Statistics
- Charts

Posts

- Manage Posts

Categories

- Manage Categories

Tags

- Manage Tags

Comments

- Moderate Comments

Users

- Manage Users

Subscribers

- Manage Subscribers

Settings

- Website Settings
-

19. Website Settings

Fields:

- Site Name
- Logo
- Favicon
- Email
- Phone
- Address
- Social Links

SEO:

- Default Meta Title
 - Default Meta Description
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20. Database Tables

Core Tables:

- users
 - roles
 - posts
 - categories
 - tags
 - post_tags
 - comments
 - subscribers
 - media
 - settings
 - contact_messages
 - post_views
-

21. Mobile Responsiveness

Support:

- Desktop
- Laptop
- Tablet
- Mobile

Framework:

- Bootstrap 5.3+
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22. Browser Compatibility

Supported Browsers:

- Chrome
 - Firefox
 - Edge
 - Safari
 - Opera
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23. Deployment Requirements

Server Requirements:

- PHP 8.3+
- MySQL 8+
- Apache/Nginx
- SSL Certificate

Deployment Commands:

- Composer Install
 - Laravel Migration
 - Laravel Cache
 - Queue Setup
-

24. Acceptance Criteria

The system shall:

- ✓ Be fully responsive
- ✓ Pass SEO audit
- ✓ Achieve Lighthouse Score above 90

- ✓ Support unlimited blog posts
- ✓ Support unlimited categories
- ✓ Support unlimited tags
- ✓ Secure user authentication
- ✓ Load pages within 2 seconds
- ✓ Operate on Laravel + MySQL production environment

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